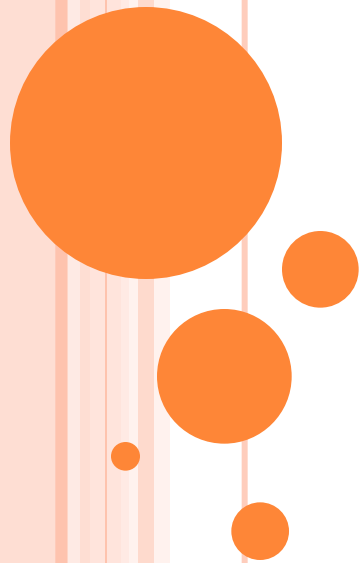


QUINE-Mc'CLUSKY'S AND MOISIL'S SIMPLIFICATION METHODS



Simplification of Boolean Functions

Informally, to simplify a Boolean function given in a disjunctive canonical form, means to cover all its minterms with a minimum number of maximal monoms and with a minimum number of overlaps.

The simplification process is formalized by the steps below.

The steps 2, 3, 4 are specific to the applied simplification method.

1. The initial function f is transformed into $DCF(f)$.
2. Factorization process $\Rightarrow$ the set of maximal monoms $M(f)$.
3. From the set of maximal monoms the central monoms are selected $\Rightarrow C(f)$
4. The case of the simplification algorithm (presented below) is identified and all simplified forms are obtained.

SIMPLIFICATION ALGORITHM

Input data: f – a Boolean function in disjunctive canonical form

Output data: $f^s_1, f^s_2, \dots, f^s_k$ – all the disjunctive simplified forms of f

begin

 Compute $M(f)$ and $C(f)$

if $M(f) = C(f)$ **then** // case 1--- one solution

$$f^s = \bigvee_{m \in M(f)} m ;$$

else

if $C(f) \neq \emptyset$ **then** // f has central monoms

 // case 2 --- k solutions

$$g = \bigvee_{m \in C(f)} m$$

$f^s_i = g \vee h_i, i = \overline{1, k}, h_i$ is a disjunction of a minimum number of maximal monoms such that $S_{h_i} = S_f \setminus S_g$

else // $C(f) = \emptyset$, no central monoms

 // case 3 --- k solutions

$f^s_i = h_i, i = \overline{1, k}, h_i$ is a disjunction of a minimum number of maximal monoms such that $S_{h_i} = S_f$

end_if

end_if

end_algorithm



QUINE'S METHOD

- it can be used for Boolean functions of any number of variables.
- it is an analitical method and it was implemented.

Definition: Let $f : (B_2)^n \rightarrow B_2$ be a Boolean function of n variables.

The set $S_f = \{(x_1, x_2, \dots, x_n) \mid f(x_1, x_2, \dots, x_n) = 1\}$, containing all the groups $(x_1, x_2, \dots, x_n) \in B_2^n$ for which f takes the value 1, is called the support of f .

Example: For $n=4$ variables,

- $m = x_1 \wedge x_2 \wedge \bar{x}_3 \wedge x_4, \quad S_m = \{(1,1,0,1)\}$
- $m = x_1 \wedge \bar{x}_3 \wedge x_4, \quad S_m = \{(1,1,0,1), (1,0,0,1)\}$
- $m = x_1 \wedge \bar{x}_3, \quad S_m = \{(1,1,0,1), (1,0,0,1), (1,1,0,0), (1,0,0,0)\}$



QUINE'S METHOD –DESCRIPTION (I)

- The Boolean function, f , is given in DCF.
- The first step is to order the support set of f , $S_f = \{(x_1, x_2, \dots, x_n) \mid f(x_1, x_2, \dots, x_n) = 1\}$, in ascending/descending order, with respect to the number of „1” values in each n-uple.
- The minterms from the function's expression are represented using the powers of variables, in a tableau, each minterm on a line, in ascending/descending order, with respect to the number of „1” values in the n-uples of the support of the function. The header of the tableau contains the variables' names.
- We make groups of minterms (delimited by horizontal lines), such that all the minterms belonging to the same group have the same number of values 1 as powers of variables. A double horizontal line marks the end of the representation of the initial function.

QUINE'S METHOD –DESCRIPTION (II)

- Only two neighbor groups can contain two adjacent (neighbor) monoms.
- The result of the factorization of two neighbor monoms (m and m') is a new monom represented as a row at the end of the tableau. The row contains the same values (0,1, or -) in the columns corresponding to the common variables (of m and m') and the symbol „-” for the variable which is eliminated. The rows corresponding to m and m' are marked (on the left side), with the meaning that they are not maximal monoms.
- All the monoms obtained as results of the factorization of two neighbor groups will form a new group (delimited by a horizontal line), used further in higher order factorizations.
- The symbol „-” cannot be combine with anything else. Thus, beginning with double factorization only rows from two neighbor groups having „-” on the same position can be combined.
- A double horizontal line symbolizes the end of a simple, double, triple,... factorization. The end of the factorization process is represented by a triple horizontal line.

QUINE'S METHOD –DESCRIPTION (III)

- The set of maximal monoms contains the monoms corresponding to all the unmarked rows from the tableau.
- For obtaining the central monoms it is used a new tableau representing the correspondence between the maximal monoms (on columns) and the minterms from the function's expression (on rows).
- A cell in the tableau is marked with a star (*) if the minterm corresponding to the row was used in factorization to obtain the maximal monom from the column. A maximal monom is a central monom if there is a * (on its column), which is unique on its row. The disjunction of all central monoms belongs to all simplified forms of the initial function.
- According to the previous tableau, the minterms from the function's expression which are uncovered by the central monoms will be covered in all possible ways using a minimum number of unused maximal monoms, with a minimum number of overlaps, resulting all simplified forms of the initial function.



EXAMPLE 1. SIMPLIFY THE BOOLEAN FUNCTION

$$f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 \vee \bar{x}_1x_2\bar{x}_3 \vee x_1\bar{x}_2\bar{x}_3 \vee x_1x_2\bar{x}_3 = m_1 \vee m_2 \vee m_3 \vee m_4.$$

$S_f = \{(0,0,1), (0,1,0), (1,0,0), (0,1,1)\}$ - support set of f

- ascending order, with respect to the number of "1" values in each triple

Group		x_1	x_2	x_3	
Representation	✓	0	0	1	m_1
I	✓	0	1	0	m_2
		1	0	0	$m_4 = \bar{x}_1\bar{x}_2x_3 = \max x_1$
II	✓	0	1	1	m_3
Simple factorizations		0	-	1	$m_1 \vee m_3 = \bar{x}_1x_3 = \max x_2$
III=I+II		0	1	-	$m_2 \vee m_3 = \bar{x}_1x_2 = \max x_3$

Maximal monoms



Example 1 (contd.)

Correspondence table:

max.monoms minterms	max_1	max_2	max_3
m_1		⊕	
m_2			⊕
m_3		*	*
m_4	⊕		

Maximal monoms:

$$max_1 = m_4 = \bar{x}_1 \bar{x}_2 \bar{x}_3$$

$$max_2 = m_1 \vee m_3 = \bar{x}_1 \bar{x}_3$$

$$max_3 = m_2 \vee m_3 = \bar{x}_1 \bar{x}_2$$

$$M(f) = \{max_1, max_2, max_3\}$$

$$= \{\bar{x}_1 \bar{x}_2 \bar{x}_3, \bar{x}_1 \bar{x}_3, \bar{x}_1 \bar{x}_2\}$$

- We circle the '*' which are unique on their rows.
- All the columns have a ⊕, thus all the maximal monoms are central monoms:

$$M(f) = C(f).$$

- The first case of the simplification algorithm => a unique disjunctive simplified form:

$$f^S(x_1, x_2, x_3) = max_1 \vee max_2 \vee max_3 = \bar{x}_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2$$



Example 2. Simplify the Boolean Function

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

The disjunctive canonical form is:

$$\begin{aligned}
 f(x_1, x_2, x_3) &= \bar{x}_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3 \vee \bar{x}_1 x_2 x_3 \vee \\
 &\quad \vee x_1 \bar{x}_2 \bar{x}_3 \vee x_1 x_2 \bar{x}_3 \vee x_1 x_2 x_3 = \\
 &= m_0 \vee m_1 \vee m_3 \vee m_4 \vee m_6 \vee m_7
 \end{aligned}$$

$$S_f = \{(0,0,0), (0,0,1), (0,1,1), (1,0,0), (1,1,0), (1,1,1)\}$$

The ascending order (with respect to the number of „1” values in each triple) of the elements of the support of f :

$$S_f = \{(0,0,0), (1,0,0), (0,0,1), (1,1,0), (0,1,1), (1,1,1)\}$$



EXAMPLE 2 – FACTORIZATION PROCESS

$S_f = \{(0,0,0), (1,0,0), (0,0,1), (1,1,0), (0,1,1), (1,1,1)\}$ - the support set of the function

Group		x_1	x_2	x_3	
I	Representation				
	✓	0	0	0	m_0
II		1	0	0	m_4
	✓	0	0	1	m_1
III		1	1	0	m_6
	✓	0	1	1	m_3
IV	✓	1	1	1	m_7
Simple factorizations V=I+II		–	0	0	$m_0 \vee m_4 = \bar{x}_2 \bar{x}_3 = \text{max1}$
		0	0	–	$m_0 \vee m_1 = \bar{x}_1 \bar{x}_2 = \text{max2}$
VI=II+III		1	–	0	$m_4 \vee m_6 = x_1 \bar{x}_3 = \text{max3}$
		0	–	1	$m_1 \vee m_3 = \bar{x}_1 x_3 = \text{max4}$
VII=III+IV		1	1	–	$m_6 \vee m_7 = x_1 x_2 = \text{max5}$
		–	1	1	$m_3 \vee m_7 = x_2 x_3 = \text{max6}$

Maximal monoms

$$M(f) = \{\bar{x}_2 \bar{x}_3, \bar{x}_1 \bar{x}_2, \bar{x}_1 \bar{x}_3, \bar{x}_1 x_3, x_1 x_2, x_2 x_3\} = \{\text{max1}, \text{max2}, \text{max3}, \text{max4}, \text{max5}, \text{max6}\}$$

□



EXAMPLE 2 (CONTD.)

Correspondence table:

max.monoms minterms	<i>max1</i>	<i>max2</i>	<i>max3</i>	<i>max4</i>	<i>max5</i>	<i>max6</i>
m_0	*	*				
m_4	*		*			
m_1		*		*		
m_6			*		*	
m_3				*		*
m_7					*	*

Factorizations:

$$m_0 \vee m_4 = \text{max1}$$

$$m_0 \vee m_1 = \text{max2}$$

$$m_4 \vee m_6 = \text{max3}$$

$$m_1 \vee m_3 = \text{max4}$$

$$m_6 \vee m_7 = \text{max5}$$

$$m_3 \vee m_7 = \text{max6}$$

There is no (*) which is unique on a certain row, each minterm is covered by two maximal monoms. $C(f)=\emptyset$, third case of simplification. Two simplified forms of f .

$$f_1^S(x_1, x_2, x_3) = \bar{x}_2\bar{x}_3 \vee \bar{x}_1x_3 \vee x_1x_2 = \text{max1} \vee \text{max4} \vee \text{max5}$$

$$f_2^S(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2 \vee x_1\bar{x}_3 \vee x_2x_3 = \text{max2} \vee \text{max3} \vee \text{max6}$$

EXAMPLE 3. SIMPLIFY THE BOOLEAN FUNCTION

$$f(x_1, x_2, x_3, x_4) = m_4 \vee m_6 \vee m_7 \vee m_8 \vee m_9 \vee m_{10} \vee m_{11} \vee m_{12}$$

- the support of f is the reunion of the supports of the minterms
- the support of the minterm m_7 is $\{(0,1,1,1)\}$
- the support of f is represented in descending order, with respect to the number of „1” values in each 4-uple

$$S_f = \{(0,1,1,1), (1,0,1,1), (0,1,1,0), (1,0,0,1), (1,0,1,0), (1,1,0,0), (0,1,0,0), (1,0,0,0)\}$$



EXAMPLE 3 - FACTORIZATION PROCESS – MAXIMAL MONOMS



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$S_f = \{(0,1,1,1), (1,0,1,1), (0,1,1,0), (1,0,0,1), (1,0,1,0), (1,1,0,0), (0,1,0,0), (1,0,0,0)\}$ Support set of f

Group		x_1	x_2	x_3	x_4	
Representation <i>I</i>	✓	0	1	1	1	m_7
	✓	1	0	1	1	m_{11}
<i>II</i>	✓	0	1	1	0	m_6
	✓	1	0	0	1	m_9
	✓	1	0	1	0	m_{10}
	✓	1	1	0	0	m_{12}
<i>III</i>	✓	0	1	0	0	m_4
	✓	1	0	0	0	m_8
Simple factorizations		0	1	1	–	$m_7 \vee m_6 = \bar{x}_1 x_2 x_3 = \max_1$
<i>IV=I+II</i>	✓	1	0	–	1	$m_{11} \vee m_9$
	✓	1	0	1	–	$m_{11} \vee m_{10}$
<i>V=II+III</i>		0	1	–	0	$m_6 \vee m_4 = \bar{x}_1 x_2 \bar{x}_4 = \max_2$
	✓	1	0	0	–	$m_9 \vee m_8$
	✓	1	0	–	0	$m_{10} \vee m_8$
		–	1	0	0	$m_{12} \vee m_4 = x_2 \bar{x}_3 \bar{x}_4 = \max_3$
Double factorizations		1	–	0	0	$m_{12} \vee m_8 = x_1 \bar{x}_3 \bar{x}_4 = \max_4$
	<i>VI=IV+V</i>	1	0	–	–	$m_{11} \vee m_9 \vee m_{10} \vee m_8 = x_1 \bar{x}_2 = \max_5$

$M(f) = \{\max_1, \max_2, \max_3, \max_4, \max_5\} = \{\bar{x}_1 x_2 x_3, \bar{x}_1 x_2 \bar{x}_4, x_2 \bar{x}_3 \bar{x}_4, x_1 \bar{x}_3 \bar{x}_4, x_1 \bar{x}_2\}$ -maximal monoms

- 4 simple factorizations and one double factorization

EXAMPLE 3 – CENTRAL MONOMS

$$M(f) = \{max_1, max_2, max_3, max_4, max_5\} = \{\bar{x}_1 x_2 x_3, \bar{x}_1 x_2 \bar{x}_4, x_2 \bar{x}_3 \bar{x}_4, x_1 \bar{x}_3 \bar{x}_4, x_1 \bar{x}_2\}$$

Correspondence table:

max.monoms minterms	max_1	max_2	max_3	max_4	max_5
m_7	⊗				
m_{11}					⊗
m_6	*	*			
m_9					⊗
m_{10}					⊗
m_{12}			*	*	
m_4		*	*		
m_8				*	*

Factorizations:

$$m_7 \vee m_6 = x_1 x_2 x_3 = max_1$$

$$m_6 \vee m_4 = \bar{x}_1 x_2 \bar{x}_4 = max_2$$

$$m_{12} \vee m_4 = x_2 \bar{x}_3 \bar{x}_4 = max_3$$

$$m_{12} \vee m_8 = x_1 \bar{x}_3 \bar{x}_4 = max_4$$

$$m_{11} \vee m_9 \vee m_{10} \vee m_8 = max_5$$

The shaded columns (max_1 and max_5) correspond to the **central monoms** because they have at least one '*', which is unique on its row.

$$C(f) = \{max_1, max_5\}$$

EXAMPLE 3 – SIMPLIFIED FORMS

$M(f) = \{max_1, max_2, max_3, max_4, max_5\} = \{\bar{x}_1\bar{x}_2x_3, \bar{x}_1x_2\bar{x}_4, x_2\bar{x}_3\bar{x}_4, x_1\bar{x}_3\bar{x}_4, x_1\bar{x}_2\}$ - maximal monoms

$C(f) = \{max_1, max_5\}$ - central monoms

$M(f) \neq C(f)$ and $C(f) \neq \emptyset$, the second case of the simplification algorithm.

$$g(x_1, x_2, x_3, x_4) \stackrel{\text{not}}{=} max_1 \vee max_5 = \bar{x}_1\bar{x}_2x_3 \vee x_1\bar{x}_2.$$

To identify the simplified forms we have to choose functions $h(x_1, x_2, x_3, x_4)$, as simple as possible, such that they cover the minterms m_{12} and m_4 uncovered by $g(x_1, x_2, x_3, x_4)$.

We choose the maximal monom having the maximum number of unshaded 'x' :

$$max_3 = x_2\bar{x}_3\bar{x}_4.$$

We remark that $h(x_1, x_2, x_3, x_4) = max_3$, covers $S_f - S_g$.

There is a unique simplified form of f , even if there is the second case of the simplification algorithm.

$$f^S(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) \vee h(x_1, x_2, x_3, x_4) = \bar{x}_1\bar{x}_2x_3 \vee x_1\bar{x}_2 \vee x_2\bar{x}_3\bar{x}_4.$$

MOISIL'S SIMPLIFICATION METHOD

- The initial Boolean function is transformed into its equivalent disjunctive canonical form.
- The set of maximal monoms are calculated using one of the previous methods (Veitch-Karnaugh diagrams, Quine's method).
- Moasil's method uses propositional logic to obtain the simplified forms from the set of maximal monoms.



EXAMPLE 4. SIMPLIFY THE BOOLEAN FUNCTION

$$\begin{aligned} f(x_1, x_2, x_3) &= \bar{x}_1\bar{x}_2\bar{x}_3 \vee \bar{x}_1\bar{x}_2x_3 \vee \bar{x}_1x_2x_3 \vee x_1\bar{x}_2\bar{x}_3 \vee x_1x_2\bar{x}_3 \vee x_1x_2x_3 = \\ &= m_0 \vee m_1 \vee m_3 \vee m_4 \vee m_6 \vee m_7 - \text{DCF (disjunctive canonical form)} \end{aligned}$$

$$S_f = \{(0,0,0), (0,0,1), (0,1,1), (1,0,0), (1,1,0), (1,1,1)\} - \text{support of } f$$

The maximal monoms were obtained using Quine's method (see Example 2).

$$M(f) = \{\bar{x}_2\bar{x}_3, \bar{x}_1\bar{x}_2, x_1\bar{x}_3, \bar{x}_1x_3, x_1x_2, x_2x_3\} = \{max_1, max_2, max_3, max_4, max_5, max_6\}.$$

Because the simplified forms of a function contain only maximal monoms we consider the following propositional sentences:

$$p_i: "max_i \text{ belongs to the simplified form of } f", i=1,2,\dots,6$$



EXAMPLE 4 (CONTD.)

p_i : " max_i belongs to the simplified form of f "

The factorization process: six simple factorizations were applied.

$$max_1 = m_0 \vee m_4 = \overline{x_2} \overline{x_3}$$

$$max_2 = m_0 \vee m_1 = \overline{x_1} \overline{x_2}$$

$$max_3 = m_4 \vee m_6 = \overline{x_1} \overline{x_3}$$

$$max_4 = m_1 \vee m_3 = \overline{x_1} \overline{x_3}$$

$$max_5 = m_6 \vee m_7 = x_1 x_2$$

$$max_6 = m_3 \vee m_7 = x_2 x_3$$

Each minterm from the function's expression must be covered by a maximal monom in a simplified form, therefore according to the result of the factorization process we have the following true sentences:

„ m_0 is covered by max_1 or by max_2 ”, translated as: $p_1 \vee p_2 \equiv T$

„ m_4 is covered by max_1 or by max_3 ”, translated as: $p_1 \vee p_3 \equiv T$

„ m_1 is covered by max_2 or by max_4 ”, translated as: $p_2 \vee p_4 \equiv T$

„ m_6 is covered by max_3 or by max_5 ”, translated as: $p_3 \vee p_5 \equiv T$

„ m_3 is covered by max_4 or by max_6 ”, translated as: $p_4 \vee p_6 \equiv T$

„ m_7 is covered by max_5 or by max_6 ”, translated as: $p_5 \vee p_6 \equiv T$



EXAMPLE 4 (CONTD.)

- All the minterms from the function's expression must be covered by a minimum number of maximal monoms, with a minimum number of overlaps. This statement is modelled by the following propositional formula obtained as a conjunction of all previous true sentences:

$$(p_1 \vee p_2) \wedge (p_1 \vee p_3) \wedge (p_2 \vee p_4) \wedge (p_3 \vee p_5) \wedge (p_4 \vee p_6) \wedge (p_5 \vee p_6) \equiv T \quad (\text{CNF with 6 clauses})$$

- The CNF (a conjunction of clauses) is transformed into DNF .
- We transform syntactically the previous formula:

$$T \equiv (p_1 \vee (p_2 \wedge p_3)) \wedge (p_4 \vee (p_2 \wedge p_6)) \wedge (p_5 \vee (p_3 \wedge p_6))$$

- Apply distributive laws:

$$T \equiv (p_1 \wedge p_4 \wedge p_5) \vee (p_1 \wedge p_4 \wedge p_3 \wedge p_6) \vee (p_1 \wedge p_2 \wedge p_6 \wedge p_5) \vee (p_1 \wedge p_2 \wedge p_6 \wedge p_3 \wedge p_6) \vee \\ (p_2 \wedge p_3 \wedge p_4 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_4 \wedge p_3 \wedge p_6) \vee (p_2 \wedge p_3 \wedge p_2 \wedge p_6 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_2 \wedge p_6 \wedge p_3 \wedge p_6)$$

- Apply idempotency:

$$T \equiv (p_1 \wedge p_4 \wedge p_5) \vee (p_1 \wedge p_4 \wedge p_3 \wedge p_6) \vee (p_1 \wedge p_2 \wedge p_6 \wedge p_5) \vee (p_1 \wedge p_2 \wedge p_3 \wedge p_6) \vee \\ \vee (p_2 \wedge p_3 \wedge p_4 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_4 \wedge p_6) \vee (p_2 \wedge p_3 \wedge p_6 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_6)$$

- Apply absorption: a DNF with 5 cubes is obtained

$$T \equiv (p_1 \wedge p_4 \wedge p_5) \vee (p_1 \wedge p_4 \wedge p_3 \wedge p_6) \vee (p_1 \wedge p_2 \wedge p_6 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_4 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_6)$$

EXAMPLE 4 (CONTD.)

p_i : " max_i belongs to the simplified form of f "

$$T \equiv (p_1 \wedge p_4 \wedge p_5) \vee (p_1 \wedge p_4 \wedge p_3 \wedge p_6) \vee (p_1 \wedge p_2 \wedge p_6 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_4 \wedge p_5) \vee (p_2 \wedge p_3 \wedge p_6) - DNF \text{ with 5 cubes}$$

A *DNF* (disjunction of cubes) is true if at least one of its cubes is true.

From *DNF* we consider the cubes with the minimum number of propositional variables and correspondingly we obtain the simplified forms of f .

For the cube $p_1 \wedge p_4 \wedge p_5 \equiv T$ the corresponding simplified form is:

$$f_1^S(x_1, x_2, x_3) = max_1 \vee max_4 \vee max_5 = \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 x_3 \vee x_1 x_2.$$

For the cube $p_2 \wedge p_3 \wedge p_6 \equiv T$, the corresponding simplified form is:

$$f_2^S(x_1, x_2, x_3) = max_2 \vee max_3 \vee max_6 = \bar{x}_1 \bar{x}_2 \vee x_1 \bar{x}_3 \vee x_2 x_3.$$

If we take the other cubes from *DNF*, there will be overlaps: a minterm will be covered by two maximal monoms and these forms do not correspond to the simplest forms of f .

